Features of the Dopaminergic Neurotoxin MPTP

IRWIN J. KOPIN
National Institutes of Health
National Institute of Neurological Disorders and Stroke
Building 10, Room 5N214
Bethesda, Maryland 20892

After the discovery that intravenous administration of 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine (MPTP) produces parkinsonism in humans and other primates, possible mechanisms of its toxicity received considerable attention.[34,38] The toxin was found to destroy specific neurons in brains of several animal species as well as in brains of primates. Vulnerability to the toxin, however, was found to vary among species (and even among strains within species), with age and with sex. In monkeys, nigrostriatal dopaminergic neurons are selectively vulnerable to the toxin, whereas in mice, which require higher doses of the toxin, the effects are less specific and involve noradrenergic, as well as other dopaminergic neurons. Rats are even more resistant to the toxin, although some effects have been reported. Although there has been enormous progress in understanding the processes involved in MPTP toxicity, a number of features remain unexplained. It is the purpose of this brief review to outline what has been learned about mechanisms involved in MPTP toxicity and to focus on several unresolved issues.

BIOACTIVATION OF MPTP IN BRAIN BY MAO-B IN ASTROCYTES

Metabolic conversion of MPTP to a toxic metabolite, 1-methyl-4-phenylpyridinium (MPP^+), is mediated by monoamine oxidase B (MAO-B). In brain, MAO-B is contained mostly in astrocytes, and MPP^+ is believed to be formed mainly in these cells. If formation of MPP^+ is blocked by inhibition of MAO-B, MPTP is not toxic. Since systemically administered MPP^+ (or MPP^+ formed in peripheral tissues) does not pass the blood–brain barrier, entry of MPTP is clearly important for its pharmacologic and toxic effects in brain.

Differences among species in MAO-B localization in brain have been cited as a basis for differing vulnerabilities to MPTP toxicity.[19,31,41] In primates, the highest levels of immunoreactive MAO-B is associated with astrocytes in the substantia nigra and basal ganglia, whereas in the rat, MAO-B is found mainly in the cells lining the ventricles and blood vessels.[41] The greater vulnerability of primate nigrostriatal neurons to MPTP toxicity may be related to the higher levels of MAO-B found in the nigrostriatal pathway. On the other hand, the high levels of MAO-B in the microvessels of rat brain have been proposed as a biochemical barrier to protect the brain of this species from circulating MPTP. Rapid conversion of lipid-soluble MPTP to MPP^+, which does not readily penetrate lipid membranes, would prevent penetration of the toxin into the brain. However, Riachi *et al.*[32] found that [^{3}H]MPTP, like [^{14}C]butanol, was extracted almost completely during a single passage of blood through the rat brain, with radioactivity distributed almost uniformly in the various areas examined. These results show that conversion of MPTP to MPP$^+$ is not sufficiently rapid to block its entry into rat brain. After entering the brain, the levels of tritium from [^{3}H]MPTP declined much more slowly than those of [^{14}C]butanol, which was rapidly washed out of the brain. Also the relatively slow decline in brain tritium could not be attributed to formation of MPP$^+$ since tritium retention was unaltered by inhibition of MAO-B. Thus, it must be concluded that MPTP not only rapidly enters, but is by some means retained in the brain without being metabolized by MAO-B. Riachi *et al.*[32] attributed retention of MPTP to its higher octanol/water partition coefficient (15.6 vs. 7.0 for butanol), but it is unlikely that this two-fold difference accounts for the greater than 15-fold difference in washout rate constants for MPTP and butanol (0.09 min^{-1} and 1.37 min^{-1}, respectively).

An alternative explanation for MPTP retention might be related to its amine moiety, which at physiological pH is almost totally protonated. Binding of cationic MPTP to acidic intracellular sites could contribute to its retention and conversion to MPP$^+$ in astrocytes. Recent studies (Marini, Schwartz and Kopin, unpublished observations) on the uptake and disposition of MPTP in cultured astrocytes support this possibility. [^{3}H]MPTP accumulates rapidly from the incubation medium into cultured rat cerebellar astrocytes, reaching concentrations much greater than those in the medium. Preincubation with pargyline has no significant effect on the astrocytic accumulation of tritium, indicating that, like retention of tritium from [^{3}H]MPTP in brain, metabolism to MPP$^+$ is not necessary for accumulation of the radioactivity.

In vitro, concentration by cells or organelles of amines from surrounding medium results, at least in part, from differences in hydrogen ion concentration. Methylamine is a weak nonbinding amine which has been used to estimate differences in pH between medium and chromaffin granules, lysosomes, chloroplasts, bacteria, etc.[17] It is assumed that only the unprotonated form of methylamine diffuses across the lipid membrane, that the concentration of the unprotonated amine in the lipid membrane-bounded compartment is identical to that in the medium, and that there is no significant binding of this simple amine. Since at physiological pH almost all methylamine is protonated, total amine concentrations closely approximate the concentrations of the amine cations. From the Henderson–Hasselbach equation it follows that the log C_{in}/C_{out} is equal to the difference between the pH in the medium and that in the organelles. When intact cells are used, if there are intracellular organelles, only an apparent difference in pH, based on the net H$^+$ concentration in the cell, can be obtained. The ratio, C_{in}/C_{out}, can, however, provide a valid estimate of amine accumulation based on differences in pH between the medium and various intracellular compartments. If cellular accumulation of an amine exceeds that of methylamine, the amine must be bound to a cellular component.

In cultured astrocytes, at physiological pH, accumulation of MPTP is about 15-fold greater than that of methylamine. MPTP is concentrated about 50-fold over the concentration in the medium, whereas methylamine is concentrated only 2.7-fold. This indicates that MPTP is bound to some astrocytic structure(s).

Many physiologically and pharmacologically active substances contain a hydrophobic ring structure and side chain with an amine group which, at physiological pH, is protonated. Because of their hydrophilic and lipophilic moieties, such cationic amphophilic compounds have complex interactions with phospholipids and phospholipid-containing membranes; they accumulate selectively in tissues, such as lung, brain and liver, which contain high levels of phospholipids.[22] Since these cationic ampholytes are bound mainly in lysosomes they are called "lysosomotropic." MPTP may be considered to be included in this class of compounds.

MPTP accumulation in cultured astrocytes is pH-dependent, and chloroquine and ammonium ions, which are highly lysosomotropic, diminish MPTP accumulation (Marini, Schwartz and Kopin, unpublished observations). Furthermore, disruption of pH gradients by monensin, a carboxylic ionophore that acts as an antiporter for sodium–hydrogen exchange,[27] decreases MPTP uptake by cultured astrocytes. Thus intra-astrocytic anionic binding sites, probably mostly lysosomes, appear to be important in the capture and retention of protonated MPTP in brain. Reversibly bound MPTP can provide a reservoir of accumulated MPTP which is sequestered until metabolized by MAO-B to 1-methyl-4-phenyl-2,3-dihydropyridinium ($MPDP^+$) and then to MPP^+.

This hypothesis explains two observations in intact animals. The first is the demonstration by Bankiewicz *et al.*[1] that unilateral intracarotid administration of MPTP to monkeys causes a contralateral hemiparkinsonian syndrome. On the side administered MPTP, dopamine levels in the basal ganglia are markedly depleted and dopaminergic neurons in the substantia nigra are almost totally obliterated. These unilateral effects are consistent with efficient extraction of MPTP from blood into brain during the first passage and rapid clearance of MPTP from the circulation. Prompt removal of MPTP from blood protects the opposite side from exposure to the toxin, whereas its retention in the exposed hemisphere allows formation of the toxic metabolite, MPP^+. Second is the report by Herkenham *et al.*[12] that MPTP-lesioned substantia nigra accumulates more [^{14}C]MPP^+ than intact substantia nigra. At 1, 3, or 10 days after administration of [^{14}C]MPTP to hemiparkinsonian monkeys (one year after they had received a toxic dose of MPTP into one carotid artery), striatal levels of [^{14}C]MPP^+ were markedly lower on the MPTP-lesioned side. This is consistent with accumulation of MPP^+ in the dopaminergic nerve terminals. The reverse was found in the substantia nigra; levels of MPP^+ were ***higher*** on the MPTP-treated side, where there was neuronal destruction and gliosis. Presumably the increased glia in the damaged substantia nigra captured the MPTP more efficiently and within one day converted it to MPP^+, which was then slowly cleared.

UPTAKE OF MPP^+ BY MONOAMINERGIC NEURONS

Specificity of MPTP toxicity for catecholaminergic neurons is conferred by the special properties of its metabolite, MPP^+, which is a substrate for the amine uptake systems in these neurons.[16] Since MPTP appears to be oxidized by MAO-B mainly in astrocytes, uptake by catecholaminergic neurons requires that the compound reach the extracellular fluid. Relatively slow appearance of MPP^+ in the cultured astrocytes and surrounding medium[35] (Marini, Schwartz and Kopin, unpublished observations) is consistent with the view that the bound, sequestered stores of MPTP provide a reservoir for cytoplasmic MPTP during its MAO-B–mediated, relatively slow, oxidation to $MPDP^+$ and MPP^+. There are several possible means by which MPP^+ appears in the extracellular fluid bathing the astrocytes. First, MPP^+ might diffuse slowly through the astrocyte cell membrane. Molecular orbital calculations imply that the positive charge of MPP^+ is highly delocalized throughout the pyridinium ring, making the compound less polar and thereby permitting slow diffusion through lipid membranes.[33] Persistence of high intracellular levels of MPP^+, however, suggests that MPP^+ does not equilibrate readily across the lipid membrane of cultured astrocytes[35] (Marini, Schwartz and Kopin, unpublished observations). Poor cell membrane penetration by MPP^+ and its very slow intracellular accumulation can also be inferred from the report by Di Monte *et al.*[7] that MPP^+ is toxic to hepatocytes in culture only after a long lag period.

Another possibility is that MPP^+ is formed extracellularly from $MPDP^+$, which diffuses out of the astrocytes. Unlike MPP^+, $MPDP^+$ can isomerize to an uncharged free base that can readily diffuse through the lipid cell membrane. When added directly to the medium, MPDP is more toxic to isolated hepatocytes than MPP^+,[8] suggesting that $MPDP^+$ enters the cells more rapidly than MPP^+. During MAO-B–catalyzed oxidation of MPTP *in vitro*, $MPDP^+$ formation is more rapid than its oxidation to MPP^+[38]; until $MPDP^+$ formation is limited by depletion of MPTP, its concentration is higher than that of MPP^+. Because of its rapid formation from MPTP and lipid-soluble properties, $MPDP^+$ might be expected to diffuse from the cytoplasm into the medium more rapidly than it is oxidized to MPP^+. Nonenzymatic conversion of $MPDP^+$ to MPP^+ (and MPTP) by disproportionation occurs at physiological pH[5] and spontaneous autoxidation of $MPDP^+$ to MPP^+ at pH 8.0 has been reported by Koytowski *et al.*[23] After addition of $MPDP^+$ to the culture medium, intracellular accumulation of MPP^+ in hepatocytes is unaffected by pretreatment with the MAO inhibitor, pargyline.[8] These results suggest that $MPDP^+$ readily diffuses into hepatocytes and that intracellular disproportionation (and perhaps spontaneous autoxidation) may be important in generating MPP^+ from $MPDP^+$. Similarly, slow diffusion of MPP^+ and more rapid diffusion of $MPDP^+$ with nonenzymatic conversion to MPP^+ may account for the appearance of the toxic metabolite in the fluid surrounding astrocytes.

VULNERABILITY TO MPP^+ TOXICITY

Uptake of MPP^+ by dopaminergic neurons via the catecholamine uptake system appears to account, at least in part, for the specificity of the toxic effects of MPTP.[16] Drugs such as mazindol, nomifensin, and amfolenic acid, which inhibit dopamine uptake, prevent MPTP toxicity in brains of mice, but it has been more difficult to demonstrate their efficacy in primates. Schultz *et al.*[37] showed that nomifensin effectively blocked MPTP neurotoxicity in primates only when administered both before, and for weeks after, the toxin, consistent with the long persistence of MPP^+ in the brain of monkeys treated with MPTP.[18] Uptake of MPP^+ does not, however, explain specific targeting of nigrostriatal dopaminergic neurons in primates. Herkenham *et al.*[12] showed that MPP^+ accumulates in many catecholaminergic neurons in monkey brain, but does not appear to permanently damage them. Thus MPP^+ toxicity appears to be related to the vulnerability of the cell rather than to accessibility of the toxin to its target.

Vulnerability of primate nigrostriatal neurons has been associated with their neuromelanin content. D'Amato *et al.*[6] reported that MPP^+ binds to neuromelanin and suggested that intraneuronal sequestration of the toxin provided a source for maintenance of cytoplasmic MPP^+ at a level that is toxic to other organelles in the neuron.

However, Herkenham *et al.*[12] found high MPP^+ levels in locus coeruleus and other monoaminergic neurons which survive its neurotoxicity. Furthermore, MPTP is toxic to sheep nigrostriatal neurons, which do not have high neuromelanin content.[10] Intracellular sequestration of MPP^+ similar to neuromelanin binding has been reported in adrenal medullary chromaffin granules, but in this case, resistance to MPP^+ toxicity was attributed to the sequestration.[33] Thus, vulnerability to MPP^+ toxicity may be related to properties of the target cell rather than to factors affecting distribution or storage of the toxin.

CELLULAR NEUROTOXICITY OF MPP^+

After entry into the cytoplasm, MPP^+ is taken up into mitochondria as a result of an energy-dependent electrochemical gradient.[30] Mitochondrial accumulation of MPP^+ attains concentrations which are sufficient to inhibit NADH dehydrogenase (Complex I) and thereby block mitochondrial respiration.[25,30] There are several important consequences of such disruption of mitochondrial energy metabolism.

Disturbance of Cytosolic Ca^{2+} Homeostasis

Cytosol concentrations of free Ca^{2+} are maintained at a level of about 100 μM, which is four orders of magnitude lower than that of extracellular Ca^{2+}. Also Ca^{2+} is sequestered in endoplasmic reticulum and in mitochondria. Expenditure of energy is required to maintain normal cytosolic Ca^{2+} concentrations. When concentrations of Ca^{2+} become elevated, cells die. Although not all of the mechanisms responsible for Ca^{2+}-mediated cell death are known, it is clear that elevation of cytosolic Ca^{2+} has been implicated in the toxicity of a wide range of toxins.[3,4]

Enhancement of Endogenous Potential Excitotoxins

ATP is also required to maintain cell membrane ionic gradients; lack of this energy source may result in a partial depolarization which could enhance vulnerability to excitotoxins. Although this may also influence cytosolic Ca^{2+}, other mechanisms may play a role. When intracellular energy levels are reduced, glutamate becomes neurotoxic, presumably as a result of relief of Mg^{2+} block of the *N*-methyl-D-aspartate receptors.[26] Turski *et al.*[40] demonstrated that selective antagonists (CPP or MK-801) of the NMDA subtype of the L-glutamate receptor, administered into the substantia nigra or systemically in repeated doses, protect against neurotoxicity of intranigral MPP^+. Preferential antagonists (CNQX and NBQX) of the quisqualate subtypes of the L-glutamate receptor failed to prevent MPP^+ toxicity. Impairment of intracellular energy metabolism and potentiation of endogenous excitotoxins has been hypothesized also by Beal *et al.*[2] to explain their observations that NMDA antagonists (MK 801 or local injection of kynurenine) prevent excitotoxic lesions produced by nigrostriatal injection of amino-oxyacetic acid. The authors suggest that AOAA blocks the pyridoxal-dependent mitochondrial malate-aspartate shunt and thereby disturbs energy production, indirectly producing excitotoxicity.

Generation of Free Radicals

Although MPP^+ redox cycling with generation of free radicals and been initially considered a possible mechanism of MPTP toxicity, one-electron reduction of MPP^+ to MPP^- is beyond the capacity of physiological reductants.[34] There is, however, evidence that free radical generation does play a role in MPTP toxicity, at least in mice. In some studies, antioxidants have conferred some protection against MPTP toxicity.[34] Recently, transgenic mice with increased superoxide dismutase have been found resistant to MPTP toxicity.[28] In another study, administration of desferrioxamine, which chelates iron and is known to diminish iron-dependent free radical damage, prevented MPTP destruction of neurons in the substantia nigra and other regions of mouse brain.[13] The mechanism of free radical generation may be related to inhibition of mitochondrial respiration. Recently, Hasagawa *et al.*[11] found that MPP^+ inhibition of bovine heart mitochondrial particle respiration generates superoxide ions (determined by the oxidation of adrenaline to adrenochrome) and enhanced NADH-supported lipid membrane oxidation. In endothelial cells, free radical formation induced by exposure to hypoxanthine and xanthine oxidase produced a sustained rise in cytosolic Ca^{2+}.[9] This effect was dependent upon extracellular Ca^{2+}. The authors suggested that lipid peroxidation and cross-linking of proteins may produce cation ionophores and transient hyperpolarization of the plasma membrane.

Disturbance of Axonal Transport

Anterograde and retrograde fast axonal transport are dependent upon ATP.[24,36] Shortly after administration of toxic doses of MPTP, the axons of the nigrostriatal neurons dorsal to the SNc become swollen and intensely fluorescent with dopamine.[15,20] These swollen regions of the axon contain MPP^+ accumulations.[12] Presumably both antero- and retrograde transport are impeded. After this effect of a toxic dose of MPTP, dopaminergic terminals appear to die or retract, after which there is a slower degeneration of the rest of the neuron. The recent discovery that brain-derived neurotrophic factor (BDNF) may support dopaminergic neuron development and diminish MPP^+ toxicity in cultured dopaminergic neurons[14,21] raises the possibility that neuronal death is a consequence of failure of retrograde axonal transport of an essential neurotrophic factor.

CONCLUSIONS

After rapid entry into the brain, retention of MPTP appears to involve sequestration in astrocyte acidic organelles. This reservoir maintains cystolic MPTP concentrations during its metabolism by MAO-B to $MPDP^+$ and MPP^+. Relatively rapid diffusion of $MPDP^+$ from the cytosol into extracellular fluids with subsequent nonenzymatic oxidation to MPP^+ and/or slower clearance of MPP^+ through the astrocyte plasma membrane makes the toxin available for uptake by monoaminergic neurons. After entry into the cytoplasm, MPP^+ is further concentrated in mitochondria, where it disrupts energy metabolism by blocking NADH dehydrogenase (Complex I). This produces an array of cellular dysfunctions; partial depolarization may enhance vulnerability to endogenous potential excitotoxins (e.g., glutamate) and/or facilitate entry of extracellular Ca^{2+}; cytosolic Ca^{2+} also may increase from mitochondrial loss of Ca^{2+}; free radical formation increases; and axonal transport is retarded. There may well be interrelationships of endogenous excitotoxins, energy depletion, free radical formation and cytosolic Ca^{2+}, each of which could account for cell damage, which proceeds to cell death, and blocking axonal transport may deprive the neuron of essential neurotrophic factors from target tissues. The peculiar vulnerability of dopaminergic nigrostriatal neurons to the toxic effects of MPP^+ remains unexplained. Most of the mechanisms of MPTP cellular toxicity have been described in tissues other than nigrostriatal dopaminergic neurons and in species other than primates. It is likely that several of these contribute to MPTP toxicity, but their relative importance may well vary with the cellular properties of the target tissue. Thus the mechanisms responsible for the death of catecholaminergic neurons in brains of mice resulting from repeated high doses of MPTP may be different from those responsible for the destruction of dopaminergic nigrostriatal neurons after a single relatively low dose of the protoxin in monkeys. The relationship of any of the factors implicated in MPTP toxicity to the pathogenesis of Parkinson's disease is tantalizing and the subject of intense study.

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